INCH-POUND
MIL-A-22641/22A
28 January 2013
SUPERSEDING
MIL-A-22641/22
21 September 1978

MILITARY SPECIFICATION SHEET

ADAPTERS, COAXIAL TO WAVEGUIDE (DOUBLE RIDGE)

(BANDWIDTH RATIO 2.4:1) (SERIES SMA TO WAVEGUIDE - CLASS 5)

INACTIVE FOR NEW DESIGN AFTER
8 MAY 1998

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the antenna described herein shall consist of this specification sheet and MIL-DTL-22641.

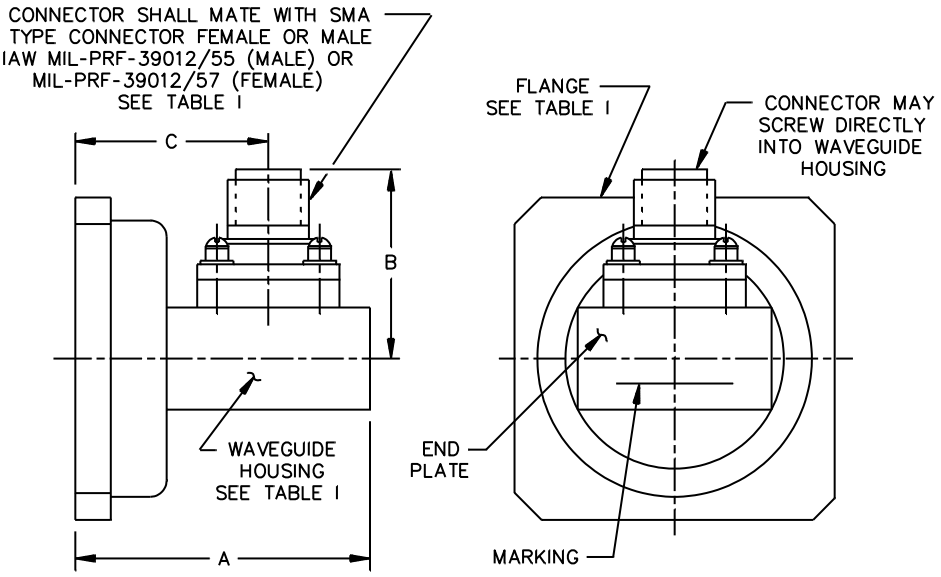


FIGURE 1. Adapters, coaxial to double ridge waveguide (class 5).

REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1 and table I.
Operating temperature range: -54º to +125º

Frequency range: See table I.
VSWR: See table I.

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TABLE I. Adapters, coaxial to waveguide (class 5 – SMA to waveguide).

Dash number	Frequency range (GHz)	Typical VSWR over any octave band with in frequency range	VSWR over entire frequency range (max) 4/ 5/	RF Insertion loss dB (max) 5/	Waveguide housing	Flange	Waveguide and flange material	Connector	Dimensions 1/ 2/ 3/		
									A (max)	B (max)	C (max)
01	.84- 2.0	1.10:1	1.15:1	.10	M23351/4-017	M39000/3-001	Aluminum	Male	4.00 (102.0)	2.20 (56.0)	2.90 (74.0)
02	.84- 2.0	1.10:1	1.15:1	.10	M23351/4-017	M39000/3-001	Aluminum	Female	4.00 (102.0)	2.20 (56.0)	2.90 (74.0)
03	2.0- 4.8	1.06:1	1.20:1	.10	M23351/4-025	M39000/3-024	Aluminum	Male	2.25 (57.2)	1.20 (30.5)	1.57 (40.0)
04	2.0- 4.8	1.06:1	1.20:1	.10	M23351/4-025	M39000/3-024	Aluminum	Female	2.25 (57.2)	1.20 (30.5)	1.57 (40.0)
05	2.0- 4.8	1.06:1	1.20:1	.10	M23351/4-025	M39000/3-026	Aluminum	Male	2.25 (57.2)	1.20 (30.5)	1.57 (40.0)
06	2.0- 4.8	1.06:1	1.20:1	.10	M23351/4-025	M39000/3-026	Aluminum	Female	2.25 (57.2)	1.20 (30.5)	1.57 (40.0)
07	2.0- 4.8	1.06:1	1.20:1	.10	M23351/4-025	M39000/3-028	Aluminum	Male	2.25 (57.2)	1.20 (30.5)	1.57 (40.0)
08	2.0- 4.8	1.06:1	1.20:1	.10	M23351/4-025	M39000/3-028	Aluminum	Female	2.25 (57.2)	1.20 (30.5)	1.57 (40.0)
09	3.5- 8.2	1.06:1	1.15:1	.10	M23351/4-029	M39000/3-030	Aluminum	Male	1.45 (37.0)	.97 (25.0)	.92 (23.0)
10	3.5- 8.2	1.06:1	1.15:1	.10	M23351/4-029	M39000/3-030	Aluminum	Female	1.45 (37.0)	.97 (25.0)	.92 (23.0)
11	3.5- 8.2	1.06:1	1.15:1	.10	M23351/4-029	M39000/3-032	Aluminum	Male	1.45 (37.0)	.97 (25.0)	.92 (23.0)
12	3.5- 8.2	1.06:1	1.15:1	.10	M23351/4-029	M39000/3-032	Aluminum	Female	1.45 (37.0)	.97 (25.0)	.92 (23.0)
13	3.5- 8.2	1.06:1	1.15:1	.10	M23351/4-029	M39000/3-034	Aluminum	Male	1.45 (37.0)	.97 (25.0)	.92 (23.0)
14	3.5- 8.2	1.06:1	1.15:1	.10	M23351/4-029	M39000/3-034	Aluminum	Female	1.45 (37.0)	.97 (25.0)	.92 (23.0)
15	4.75- 11.0	1.06:1	1.15:1	.10	M23351/4-033	M39000/3-036	Aluminum	Male	1.75 (44.5)	.90 (23.0)	1.20 (30.5)
16	4.75- 11.0	1.06:1	1.15:1	.10	M23351/4-033	M39000/3-036	Aluminum	Female	1.75 (44.5)	.90 (23.0)	1.20 (30.5)
17	4.75- 11.0	1.06:1	1.15:1	.10	M23351/4-033	M39000/3-038	Aluminum	Male	1.75 (44.5)	.90 (23.0)	1.20 (30.5)
18	4.75- 11.0	1.06:1	1.15:1	.10	M23351/4-033	M39000/3-038	Aluminum	Female	1.75 (44.5)	.90 (23.0)	1.20 (30.5)
19	4.75- 11.0	1.06:1	1.15:1	.10	M23351/4-033	M39000/3-040	Aluminum	Male	1.75 (44.5)	.90 (23.0)	1.20 (30.5)
20	4.75- 11.0	1.06:1	1.15:1	.10	M23351/4-033	M39000/3-040	Aluminum	Female	1.75 (44.5)	.90 (23.0)	1.20 (30.5)
21	7.5- 18.0	1.10:1	1.20:1	.10	M23351/4-037	M39000/3-072	Aluminum	Male	1.30 (33.0)	.75 (19.0)	.77 (20.0)
22	7.5- 18.0	1.10:1	1.20:1	.10	M23351/4-037	M39000/3-072	Aluminum	Female	1.30 (33.0)	.75 (19.0)	.77 (20.0)
23	7.5- 18.0	1.10:1	1.20:1	.10	M23351/4-037	M39000/3-074	Aluminum	Male	1.30 (33.0)	.75 (19.0)	.77 (20.0)
24	7.5- 18.0	1.10:1	1.20:1	.10	M23351/4-037	M39000/3-074	Aluminum	Female	1.30 (33.0)	.75 (19.0)	.77 (20.0)
25	7.5- 18.0	1.10:1	1.20:1	.10	M23351/4-037	M39000/3-076	Aluminum	Male	1.30 (33.0)	.75 (19.0)	.77 (20.0)
26	7.5- 18.0	1.10:1	1.20:1	.10	M23351/4-037	M39000/3-076	Aluminum	Female	1.30 (33.0)	.75 (19.0)	.77 (20.0)

1/ Dimensions are in inches.
2/ Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3/ Metric equivalents are in parentheses.
4/ A larger attenuator pad may be used for VSWR test setup (see figure 1 of MIL-DTL-22641)
5/ VSWR and RF insertion loss may be measured by swept frequency techniques.

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Referenced documents. In addition to MIL-DTL-22641, this document references the following:
MIL-W-23351/4
MIL-F-39000/3
MIL-PRF-39012/55
MIL-PRF-39012/57

The margins of this specification sheet are marked with vertical lines to indicate where modifications from this revision were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Preparing activity:
DLA – CC
(Project 5985-2013-011)

Review activities:
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Navy – AS, MC, OS
Air Force – 99

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